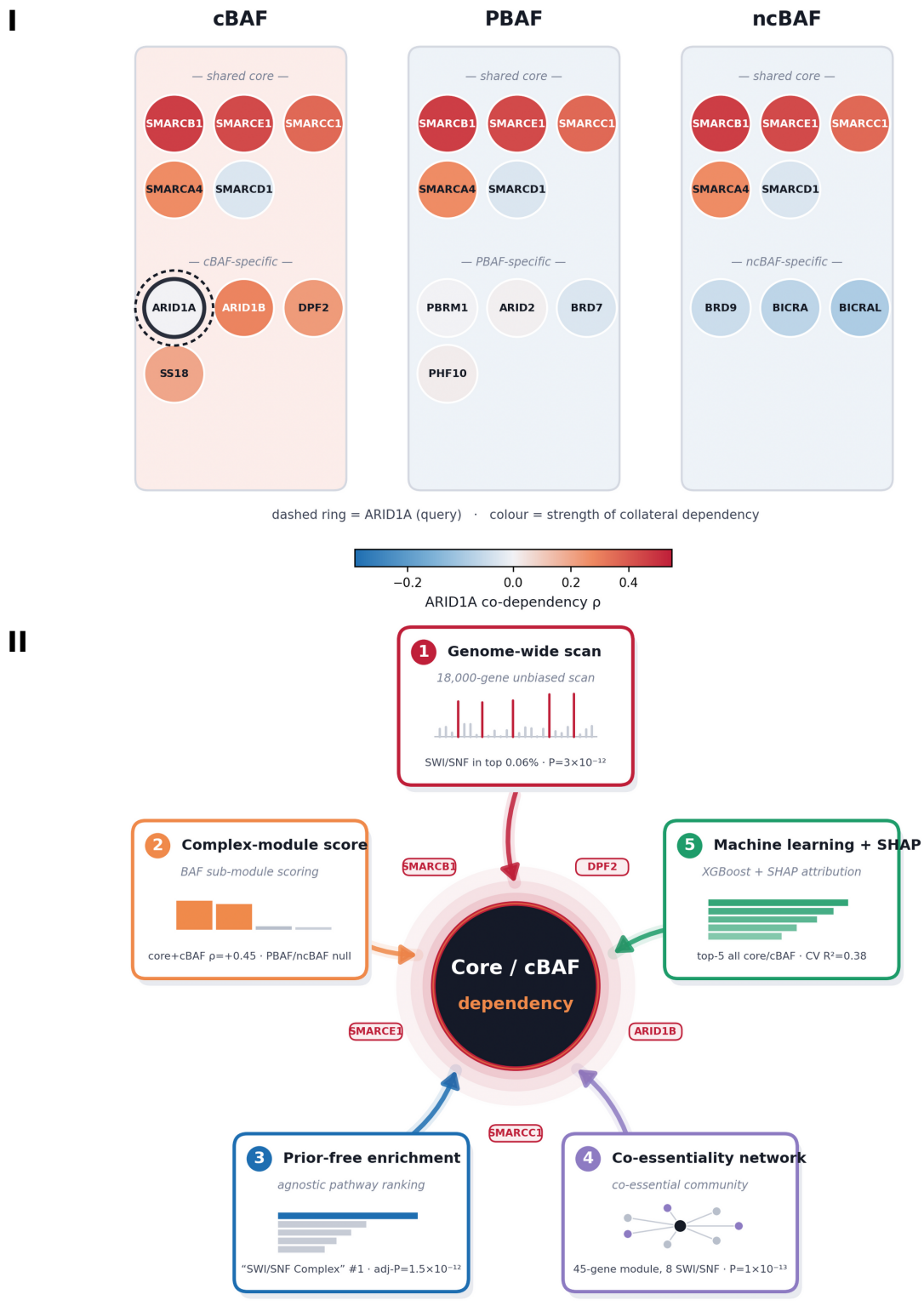
Supplementary material

Structure-informed CRISPR co-dependency maps the ARID1A-associated BAF module across cancer cell lines

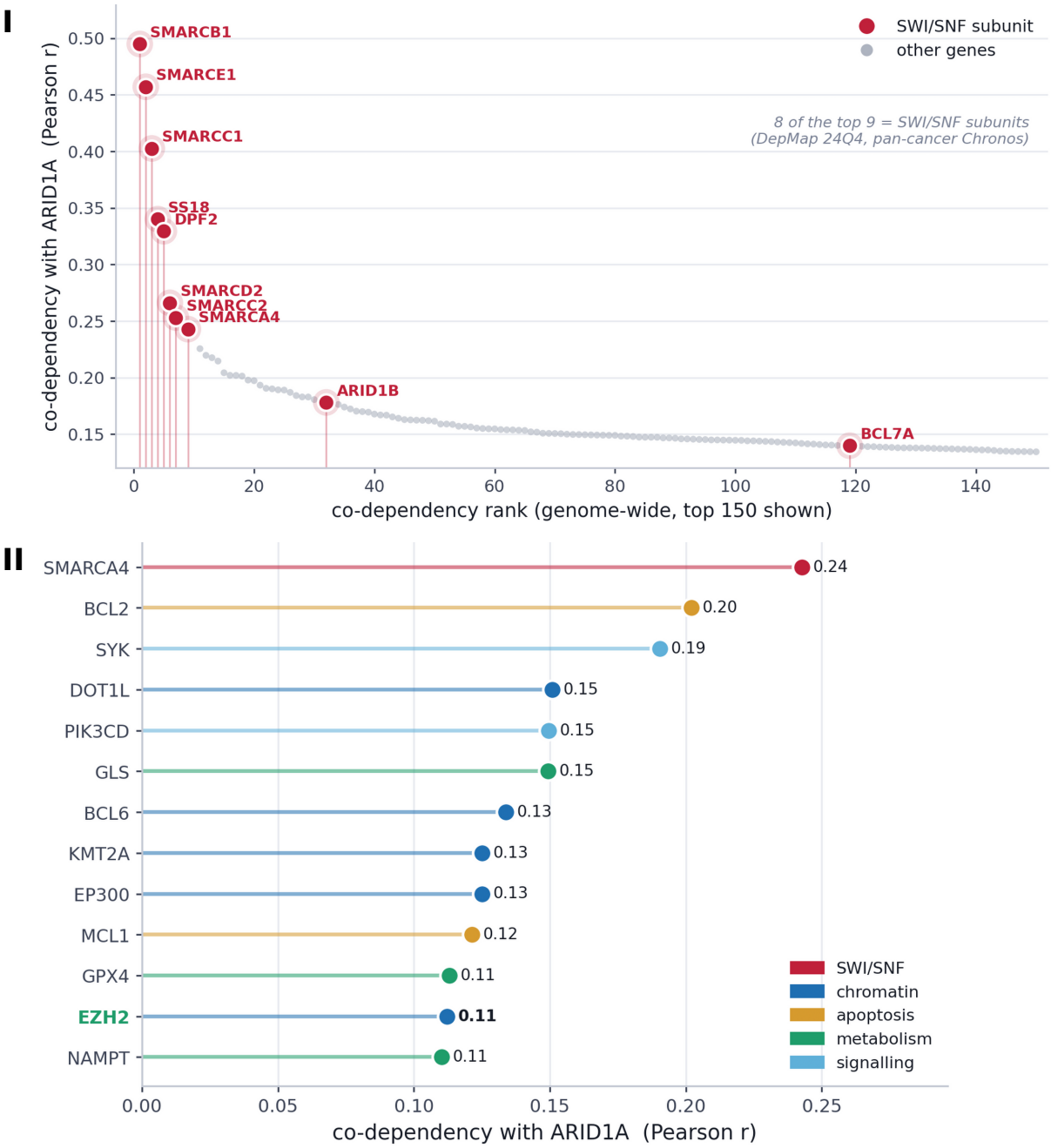
Zhiyong Tao, Juan Xiong, Yiwei Zhang, Chulin Zhong, Wenkai Gao, Guohua Fan

Supplementary Figure S1



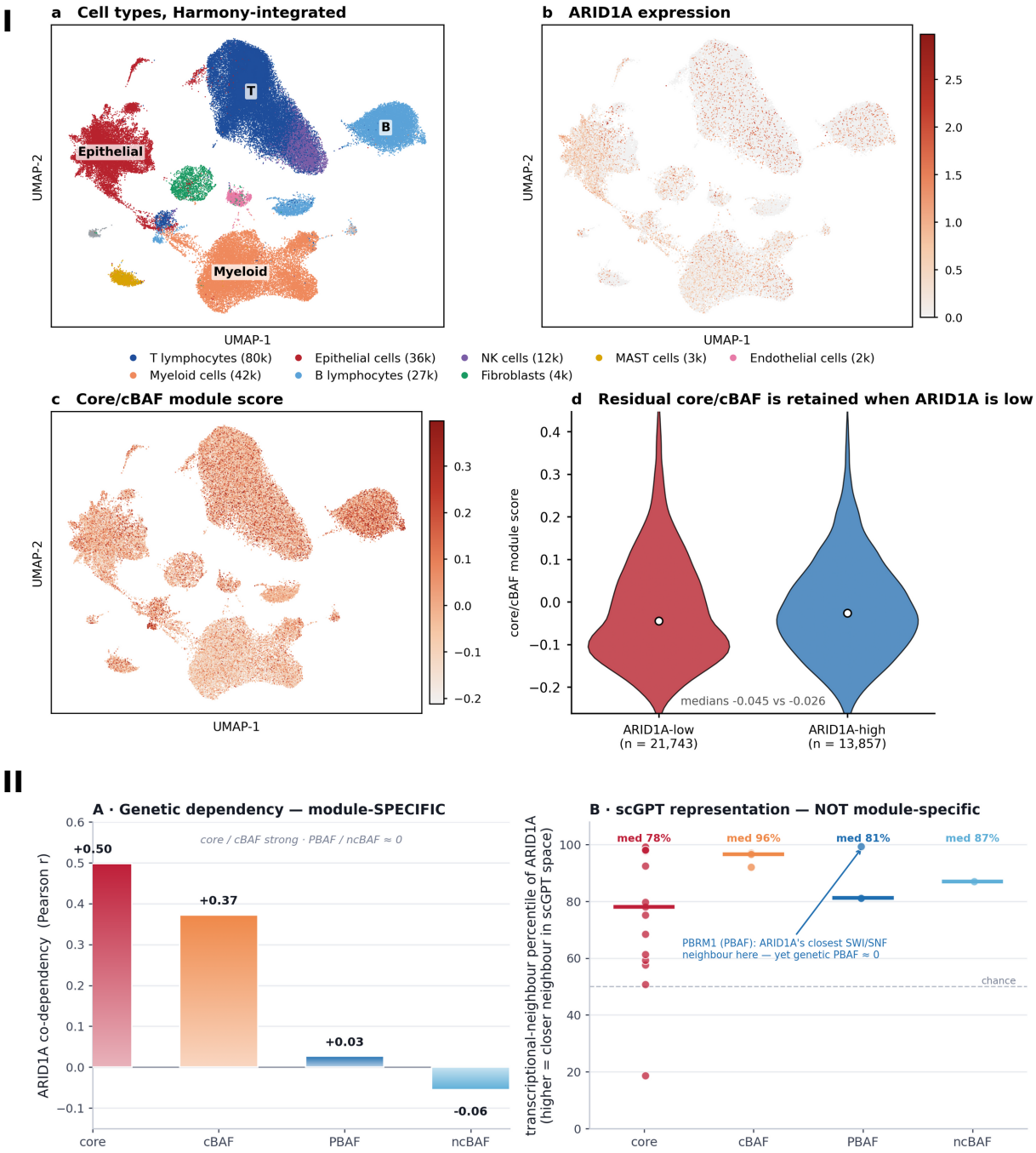
Sub-complex composition and the structure of the convergence argument. (I) The three SWI/SNF sub-complexes drawn by subunit membership. Each subunit is coloured by its pan-cancer co-dependency with ARID1A; ARID1A carries a dashed ring. (II) The five methodologically distinct analyses and the subunits each recovers. The analyses share one dependency matrix; four use no complex composition during discovery.

Supplementary Figure S2



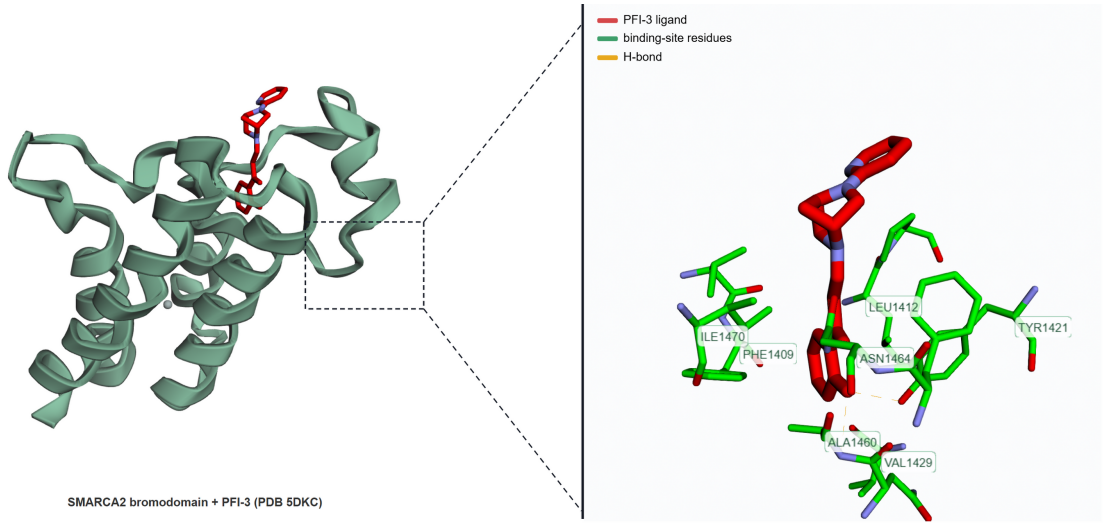
Network-guided druggability prioritisation from the genome-wide ARID1A co-dependency landscape. (I) Top 150 genes ranked by pan-cancer co-dependency in DepMap 24Q4. (II) Co-dependent genes for which a chemical probe or clinical-stage inhibitor exists. This is a co-dependency-network overlay, not a compound–target–pathway network-pharmacology analysis. The rankings prioritise follow-up targets and do not establish genotype-selective efficacy.

Supplementary Figure S3



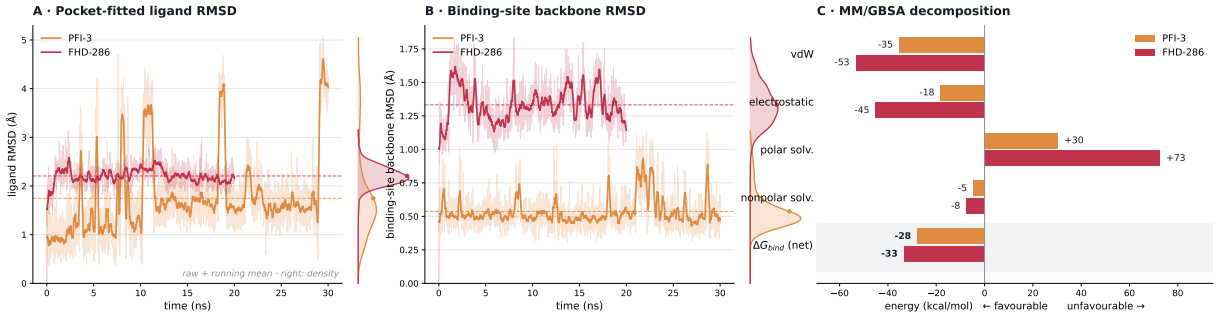
Single-cell expression context and foundation-model control. (I) GSE131907 cell types, ARID1A expression, core/cBAF module score and the descriptive cell-level split. Statistical inference in the main text uses 48 sample-level aggregates. (II) CRISPR co-dependency separates core/cBAF from PBAF/ncBAF, whereas the scGPT transcriptional representation places every module above chance and therefore does not resolve genetic specificity.

Supplementary Figure S4



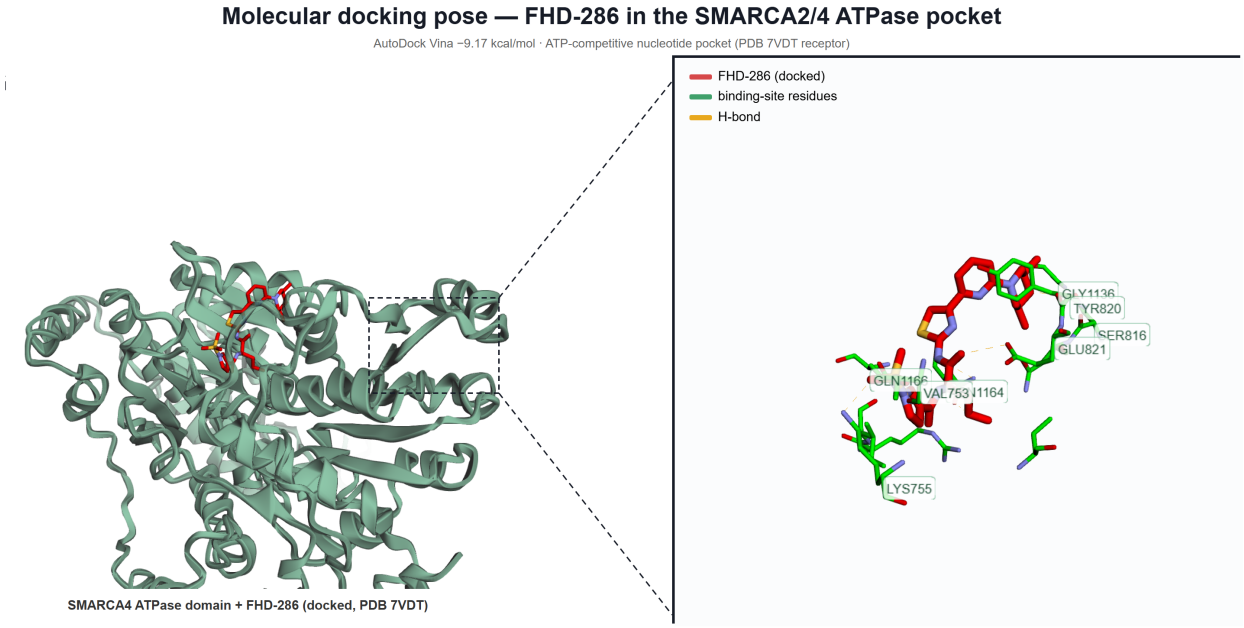
Crystallographic PFI-3 pose used as the docking positive control. Left, the SMARCA2/4 bromodomain co-crystal structure (PDB 5DKC) with PFI-3 shown in red. Right, the binding-site close-up with contacting residues labelled and the interaction with ASN1464 indicated. AutoDock Vina redocking reproduced the deposited ligand pose at 0.48 Å heavy-atom RMSD. This panel validates recovery of one known pose under the protocol; it does not validate the FHD-286 blind-docking pose or establish genotype-selective activity.

Supplementary Figure S5



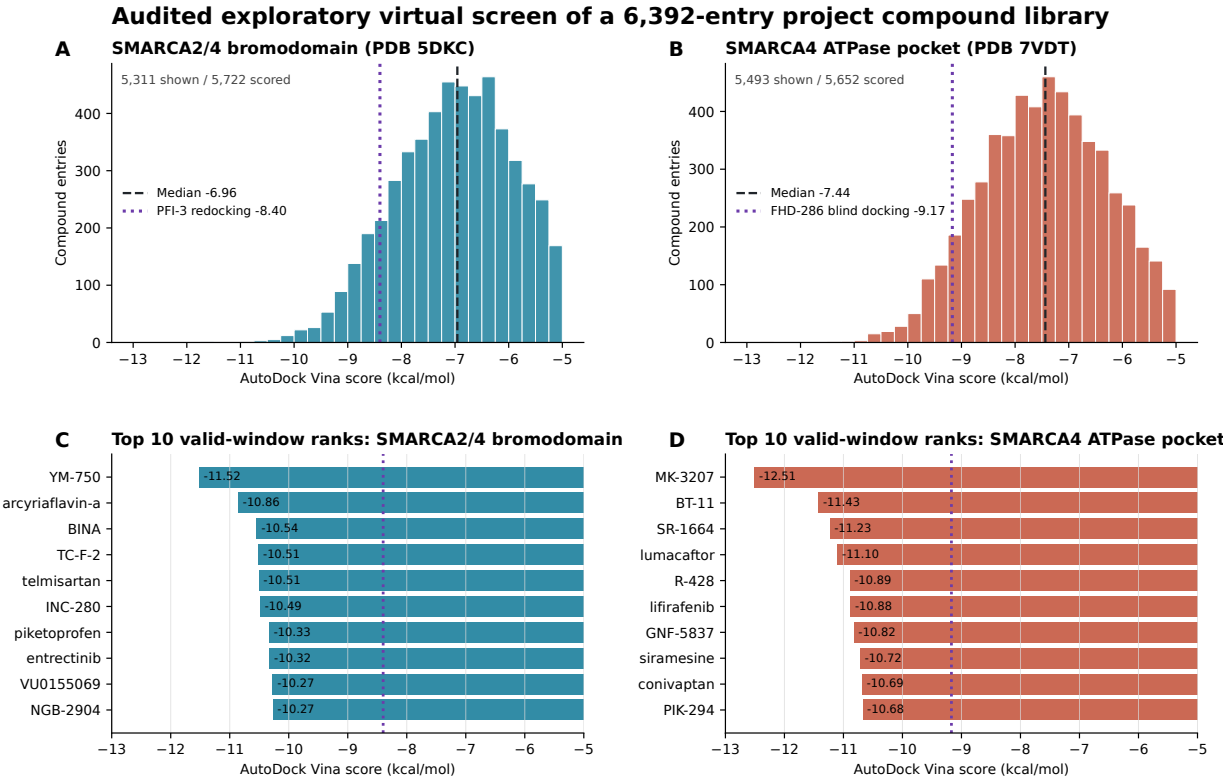
Single-trajectory molecular-dynamics checks of modelled SWI/SNF ligand poses. (**A**) Ligand RMSD after fitting the binding-site residues, shown as raw frames, running means and framewise distributions. (**B**) Binding-site backbone RMSD. PFI-3 was followed for 30 ns from its 5DKC crystallographic pose; FHD-286 was followed for 20 ns from the blind-docked 7VDT pose. (**C**) Single-trajectory MM/GBSA energy decomposition. The reported means and standard deviations describe frames within one trajectory per complex and are not uncertainty across independent simulations. These calculations assess bounded pose behaviour over the sampled windows; they do not validate the FHD-286 pose, provide absolute binding affinities or establish ARID1A-genotype selectivity.

Supplementary Figure S6



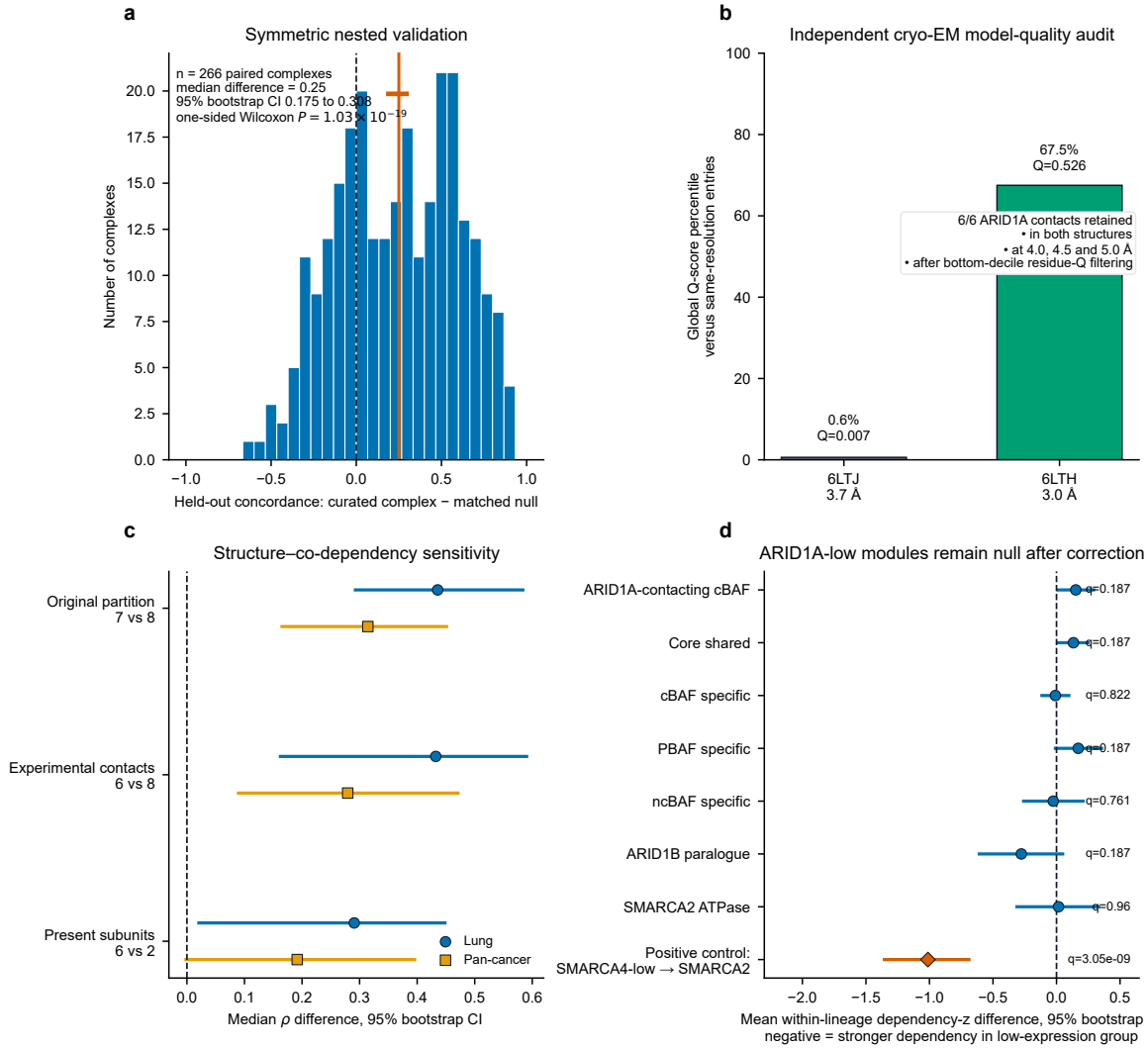
Blind-docked FHD-286 pose in the SMARCA4 ATPase nucleotide pocket. Whole-domain and close-up views show the modelled FHD-286 pose and nearby pocket residues in PDB 7VDT; the AutoDock Vina score was $-9.17\,\mathrm{kcal\,mol^{-1}}$. Unlike the crystallographic PFI-3 positive control in Supplementary Figure S4, no public co-crystal validates this pose. Residue labels therefore describe the modelled geometry and not experimentally established contacts, affinity or ARID1A-genotype selectivity.

Supplementary Figure S7



Audited exploratory virtual screen against the two modelled SWI/SNF pockets. (**A**, **B**) Score distributions for the SMARCA2/4 bromodomain (PDB 5DKC) and SMARCA4 ATPase pocket (PDB 7VDT), respectively; dotted lines mark the PFI-3 and FHD-286 reference docking scores. Of 6,392 library entries, 5,722 and 5,652 returned numeric scores, of which 5,311 and 5,493 fell inside the project's -13 to $-5\,\mathrm{kcal\,mol^{-1}}$ artefact-exclusion window. (**C**, **D**) Ten lowest scores in that window for each pocket after collapsing duplicate identifiers to their lowest score. The archived table combines an initial Vina run at exhaustiveness 8 and a resumed run at exhaustiveness 4; exact library source/version metadata were not retained. The display window is not an affinity threshold, and the rankings are hypothesis-generating rather than evidence of binding or repurposing readiness.

Supplementary Figure S8



Robustness analyses for nested module refinement, structural contact and conditional dependency. (**a**) For 266 curated complexes, the histogram shows the paired difference in held-out drop-identity concordance between each real complex and its size-matched null modules. Anchor and dropped member were selected only in the discovery half in both arms, then evaluated unchanged in the validation half over 30 splits. The orange line is the paired median difference (0.25) and the horizontal orange segment its 95% percentile bootstrap interval (0.175–0.308); the dashed line marks zero. (**b**) wwPDB global Q-score percentiles relative to entries at the same resolution. The full 6LTJ model is low quality globally, whereas the independent 6LTH base-module deposit is above the median; nevertheless, all six binary ARID1A contact calls persist in both deposits, across 4.0–5.0 Å cutoffs and after bottom-decile residue-Q filtering. (**c**) Median co-dependency difference between contacting and non-contacting groups under three structural definitions, with 95% bootstrap intervals. Lung and pan-cancer estimates are shown separately; exact two-sided Mann–Whitney tests are significant for the original 7-vs-8 and experimental-contact-only 6-vs-8 definitions ($P \leq 0.0127$), whereas the 6-vs-2 restriction is underpowered ($P = 0.286$). (**d**) Mean within-lineage dependency-z difference for the bottom 10% of ARID1A protein abundance versus the remainder, with 95% bootstrap intervals and BH-adjusted q values (52 low and 483 remaining lines). No module passes correction. The prespecified pipeline control applies the same design to the bottom 10% of SMARCA4 mRNA and detects stronger SMARCA2 dependency (102 low lines; $\Delta z = -1.012$, 95% interval -1.355 to -0.689; $q = 3.05 \times 10^{-9}$). Intervals use 20,000 bootstrap resamples with seed 20260901.